TEL200 Introduction to Robotics

Robot vision & learning

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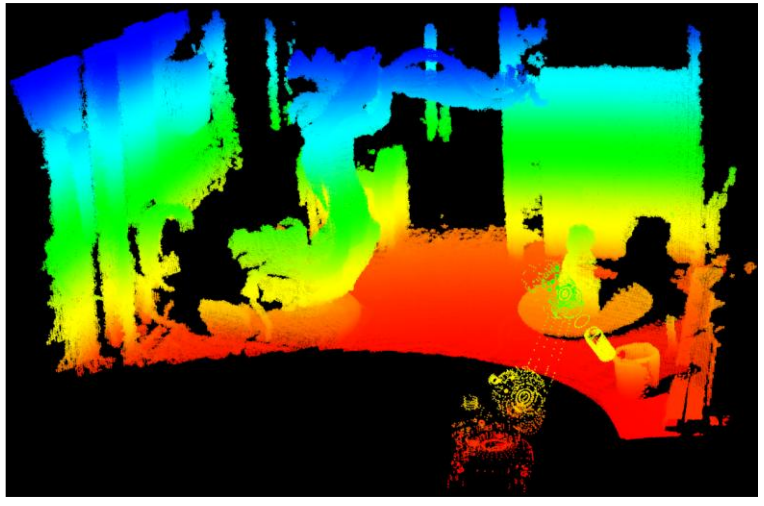
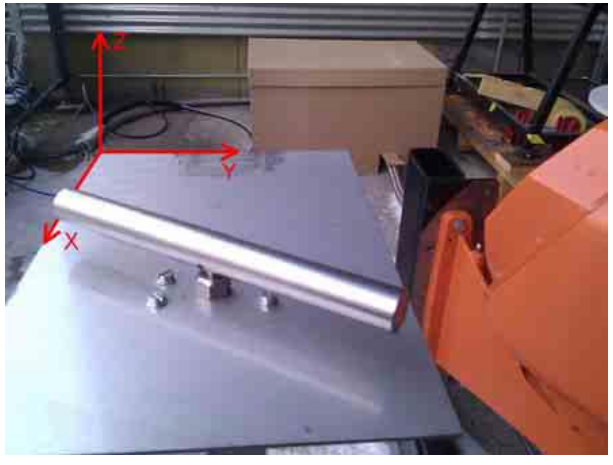
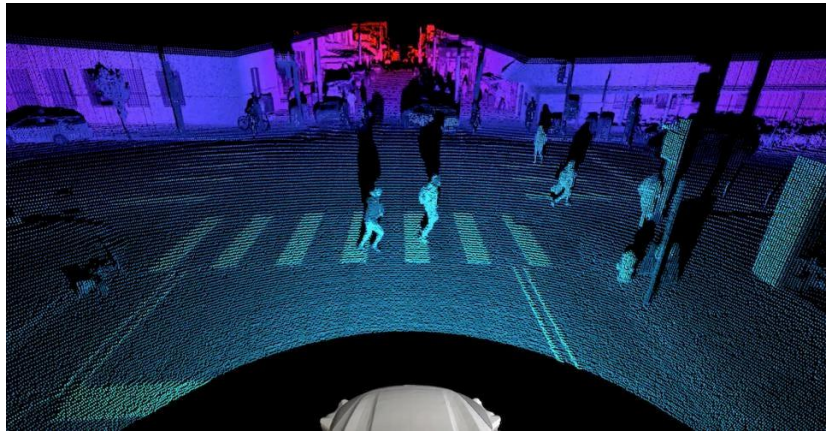
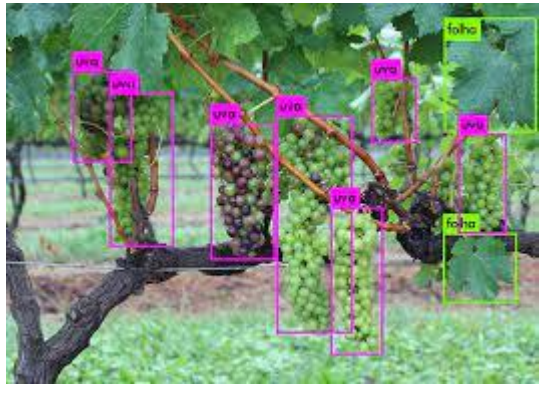
Robot vision

- Automatic extraction of meaningful information from images and videos



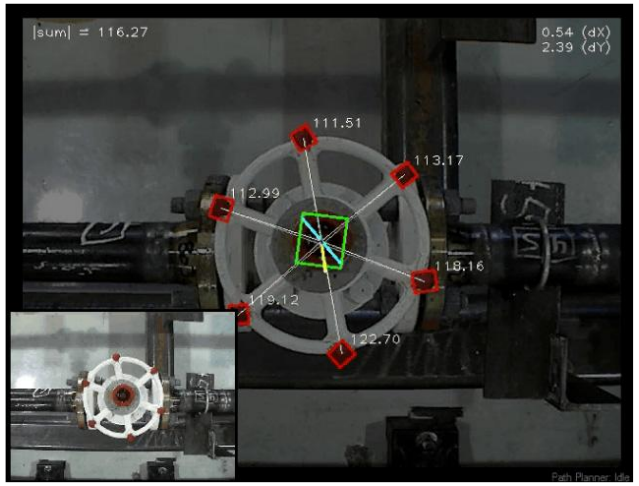






Robot vision - Challenges

- Illumination
- Clutter
- Occlusions
- Viewpoint
- Complexity



Robot/Machine vision - Topics

Low-Level

- Texture Analysis
- Edge Detection
- Optical flow estimation
- Image matching
- Corner Detection
- Blob Detection

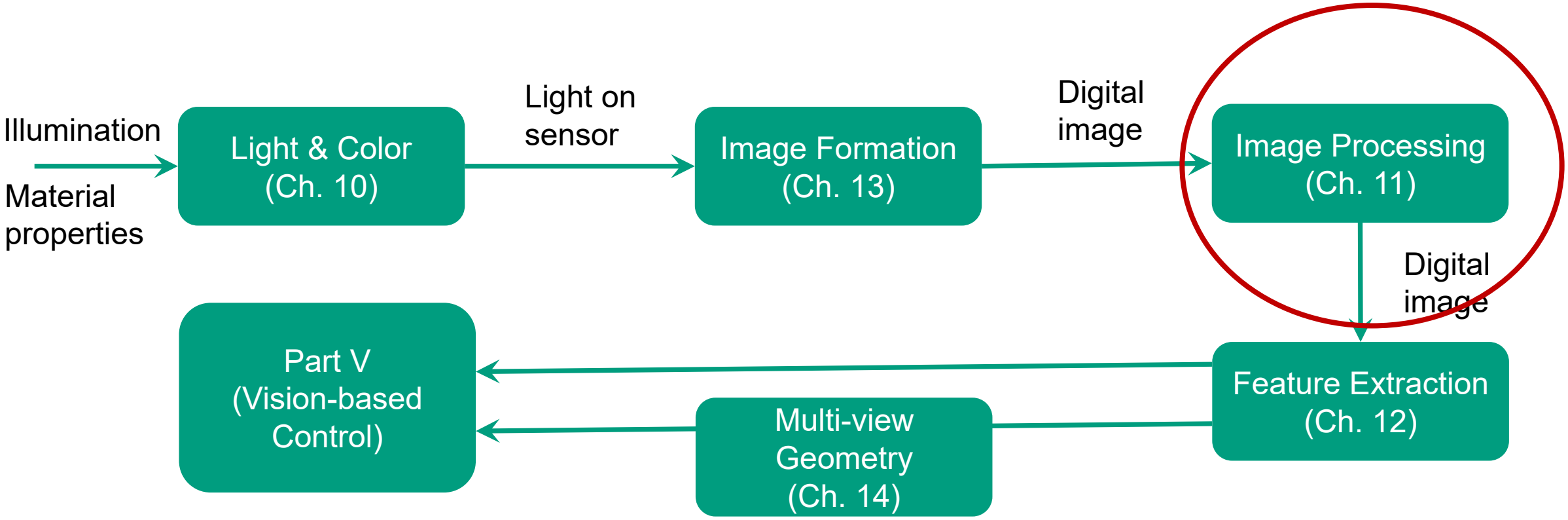
Middle-Level

- Geometry analysis
- Multi-View geometry
- Structure from motion
- Stereo analysis
- Segmentation
- Visual tracking
- Motion analysis

High-Level

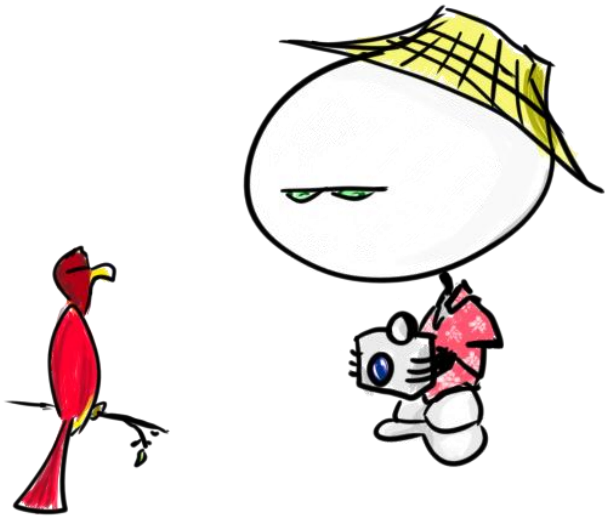
- Semantics
- Object Recognition
- Scene Understanding
- Image understanding
- Video understanding
- Machine Learning

Machine vision in RVC, Ed 3

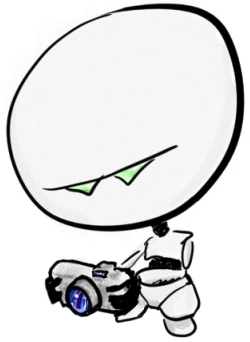


Agenda

- Why use camera on a robot?
- Ch. 11.1.1 «Images from Files»
  - Digital pictures
- Ch. 10.2.7 Color-spaces (fargemodeller)
- Ch. 11.3 Monadic operations (punkteoperasjoner)
- Ch. 11.5 Spatial operations (maskeoperasjoner)



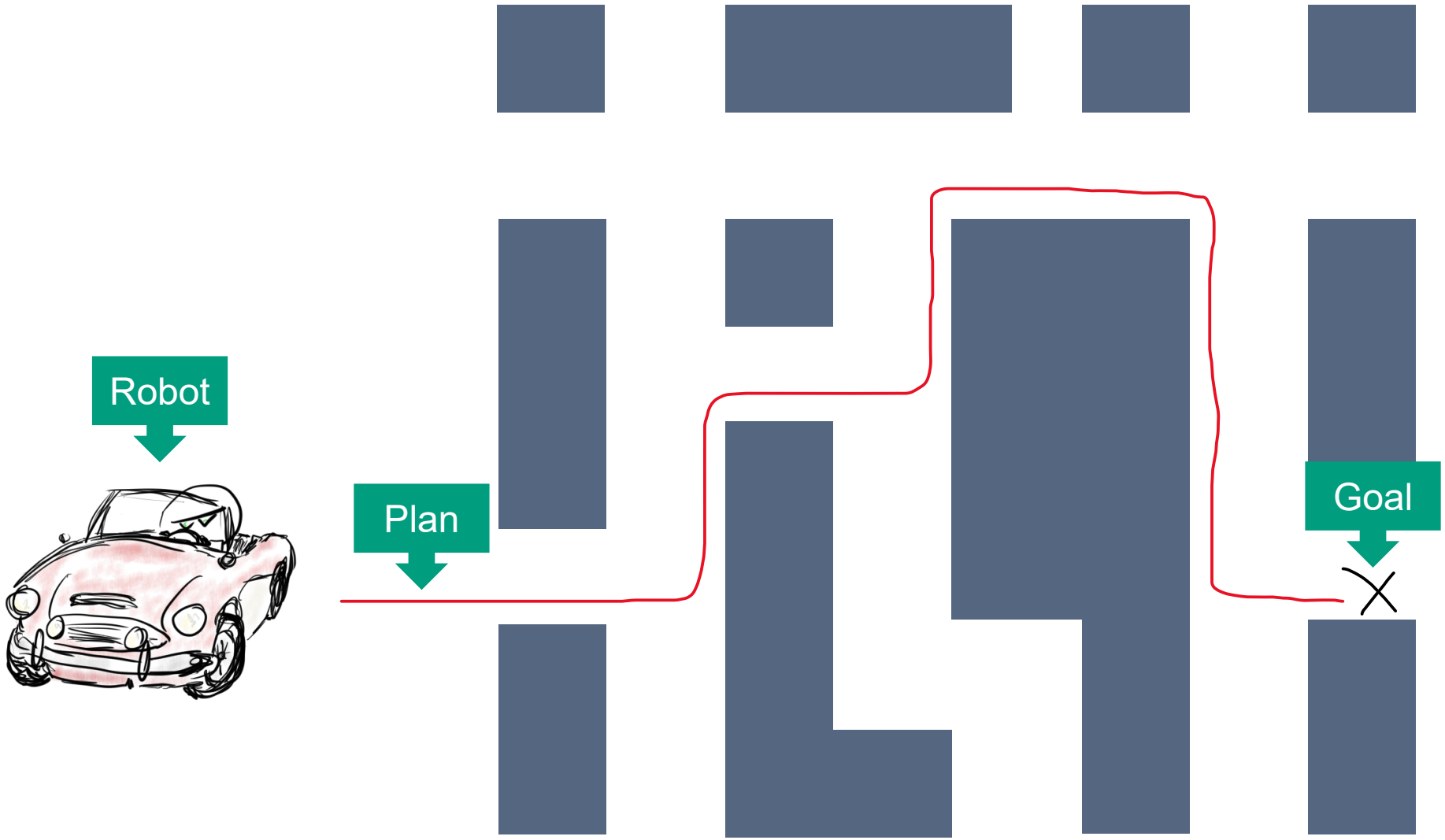
Why use camera on a robot?



Robot: Goal-oriented machine that can sense, plan and act

Robot: Goal-oriented machine that can sense, plan and act

Where am I?
What is the surrounding environment?
How to reach the gaol?

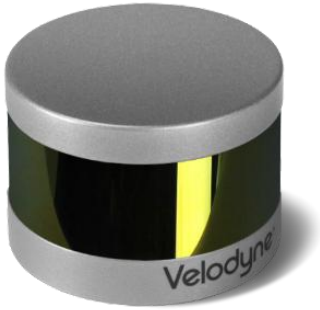

Robot
Plan
Goal

UWB radar
(e.g. Novelda)

GNSS

IMU

Stereo-camera



3D-lidar

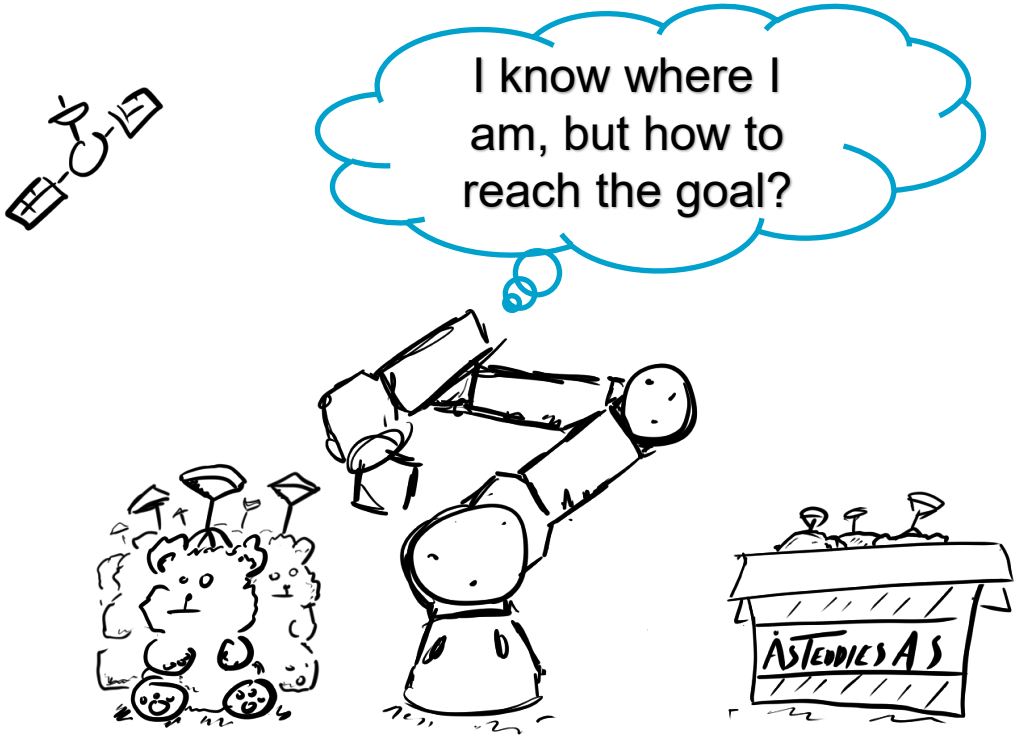
Ultrasound

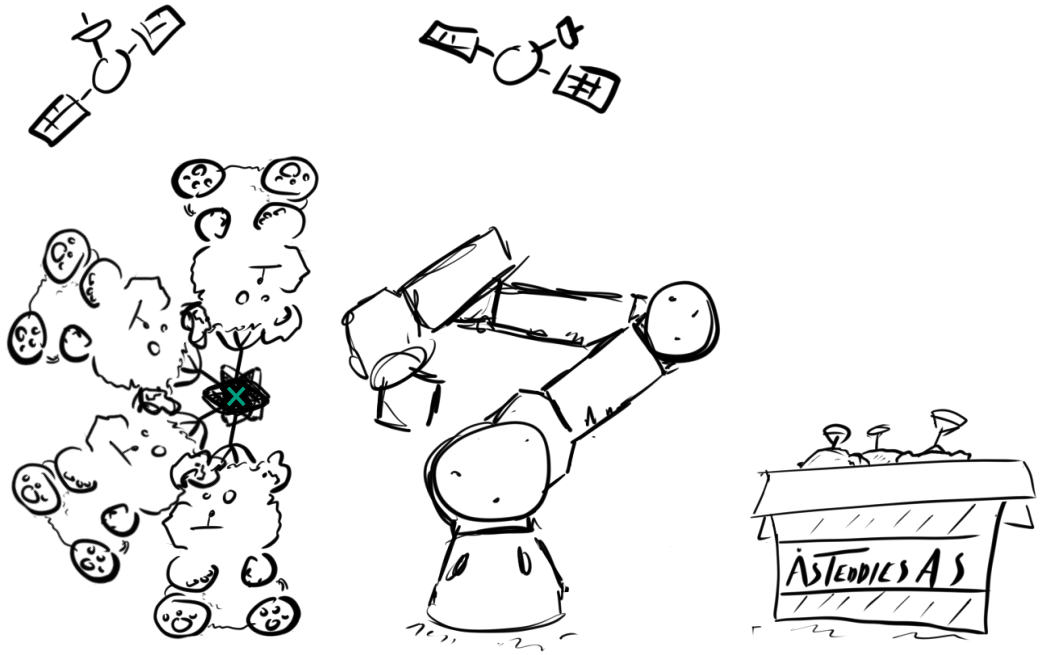
Camera

2D-lidar

GNSS

- GNSS is often useful, but seldom sufficient in its own
  - Poor satellite coverage in parts of city, forest, valleys, etc.
  - Cannot equip everything with GNSS-receivers
- Need to «see» the environment: Lidar, radar, camera


I know where I am, but how to reach the goal?


ÅSTEDILSAS

Related QUT Robot Academy video:

- Introduction to robotic vision:
  - https://robotacademy.net.au/masterclass/robotic-vision/?lesson=589
- Robots revisited:
  - https://robotacademy.net.au/masterclass/robotic-vision/?lesson=592
- Why not use GPS:
  - https://robotacademy.net.au/masterclass/robotic-vision/?lesson=596

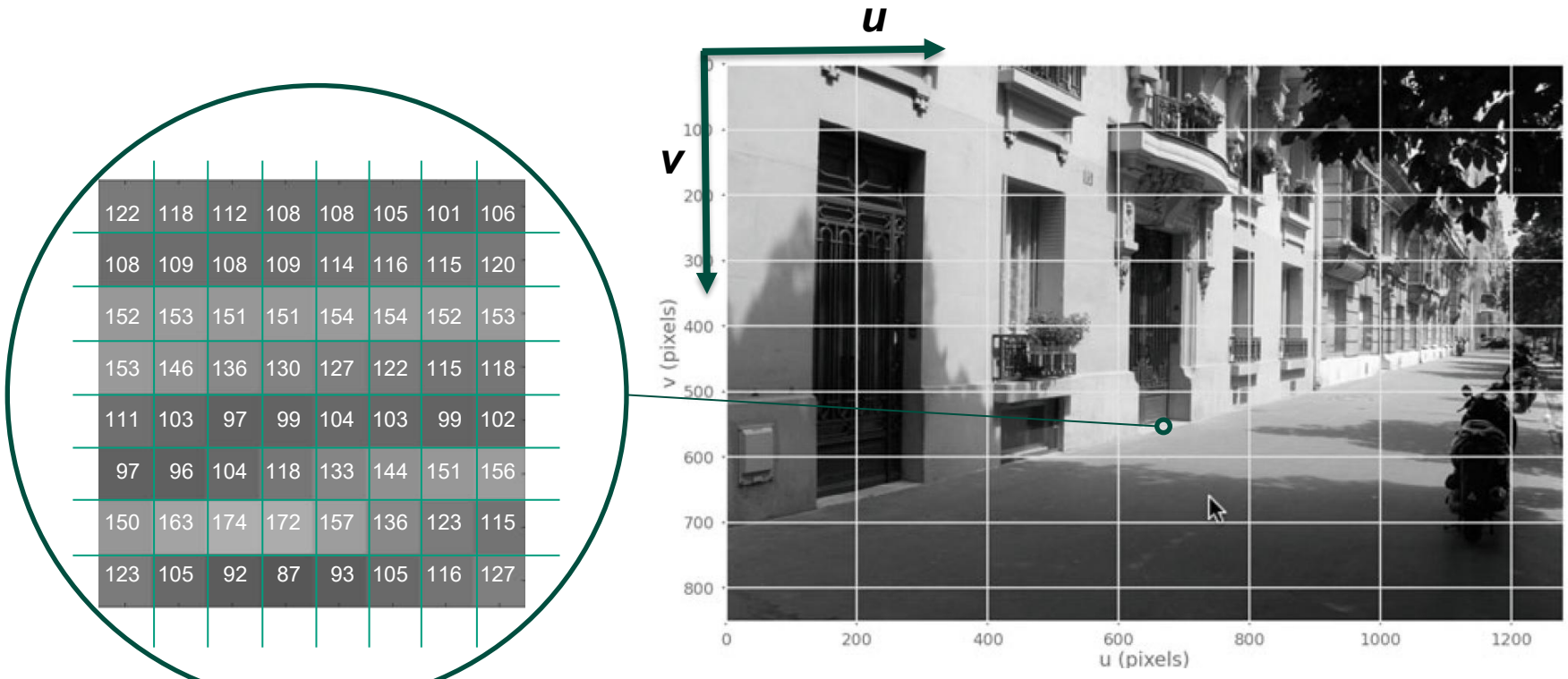
11.1.1 Images from Files

Digital images

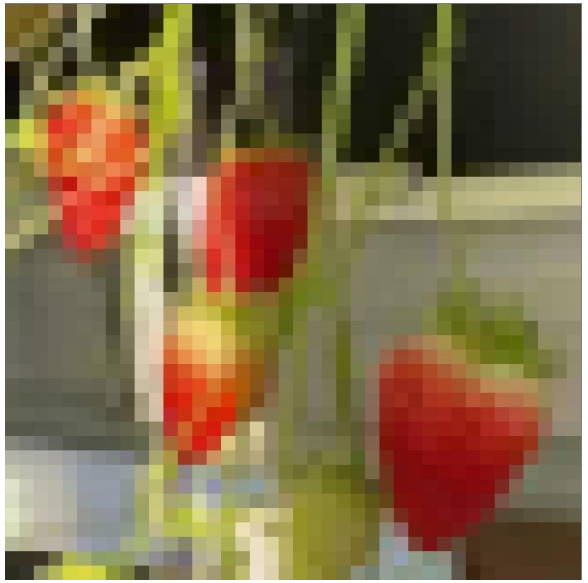
- Pixels, smallest entity in pictures
- 8 bit (1 byte), unsigned int → intensity 0 – 255
- «Double precision» → 8 byte/piksel → intensity 0.0 – 1.0 (typical)

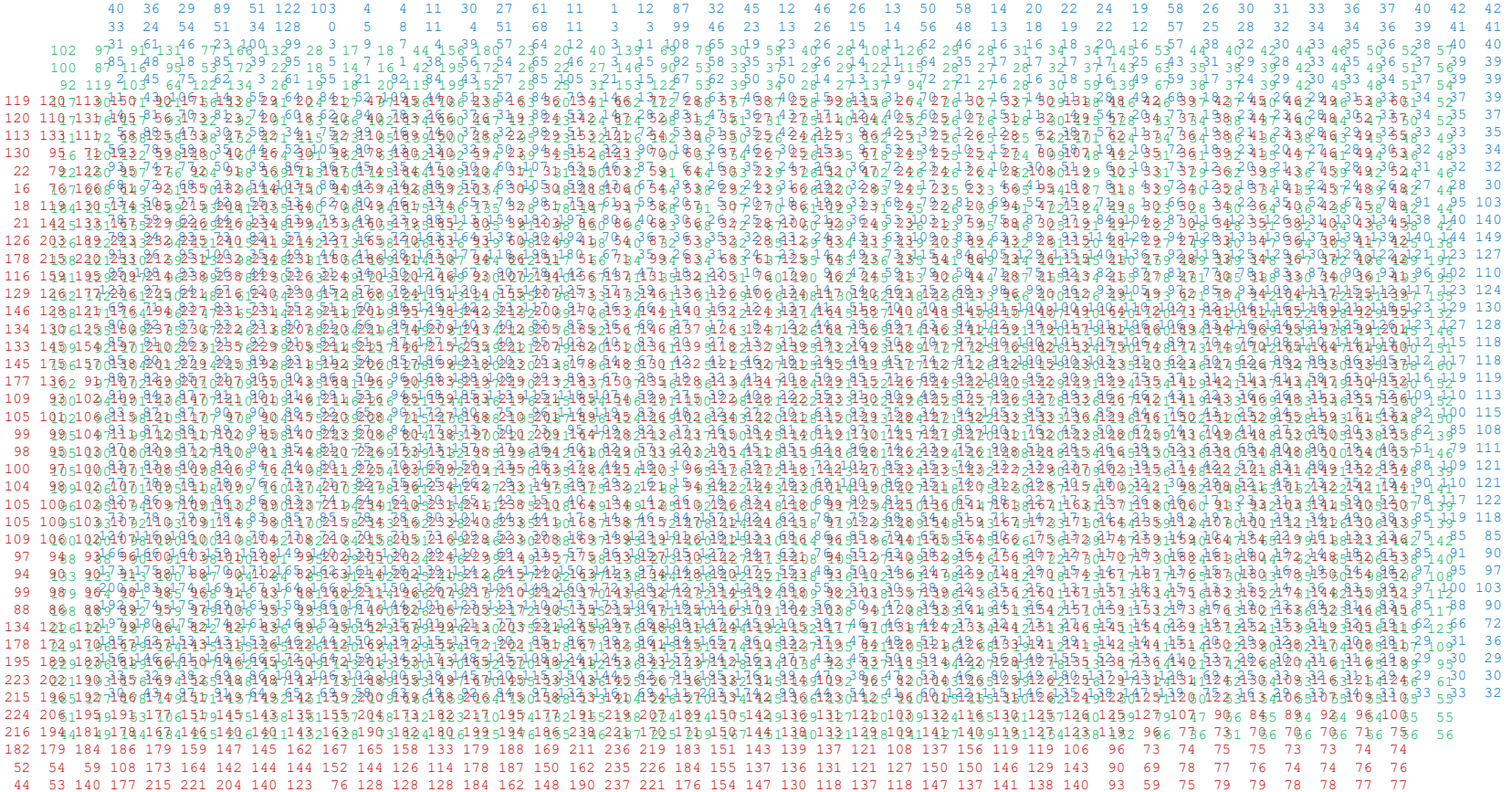
```
>>> street, _ = iread("street.png");
>>> street
array([[193, 193, 191, ..., 8, 7, 6],
[189, 190, 191, ..., 7, 7, 8],
[190, 189, 190, ..., 8, 8, 7],
...,
[ 50, 51, 52, ..., 30, 28, 29],
[ 53, 54, 53, ..., 28, 28, 29],
[ 53, 56, 54, ..., 27, 27, 28]], dtype=uint8)

>>> idisp(street);
```



Color image – RGB

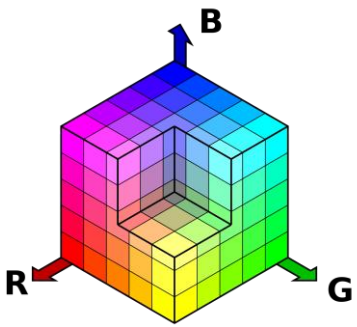




Color image – RGB

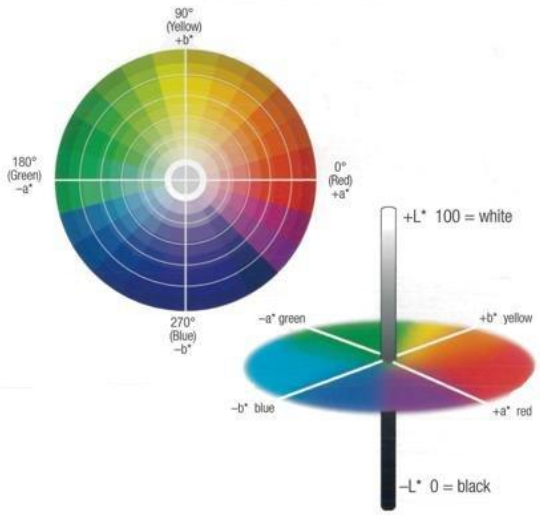
R (red) *G* (green) *B* (blue)

(row, column, color space)



Other color spaces (Ch. 10.2.7) - L*a*b (CIELAB)

*L** (Lightness) *a** (green-red) *b** (blue-yellow)



Related QUT Robot Academy video:

- Getting an image into a computer
  - https://robotacademy.net.au/masterclass/getting-images-into-a-computer/?lesson=628
- Reading an image from a local camera
  - https://robotacademy.net.au/masterclass/getting-images-into-a-computer/?lesson=631
- Summary of digital images
  - https://robotacademy.net.au/masterclass/getting-images-into-a-computer/?lesson=635

11.3 Monadic Operations (punktoperasjoner)

- Each pixel in the output image is a function of corresponding pixel in original/input image
- Examples:
  - Convert a picture fra uint8 to double
  - Convert a picture fra double to uint8
  - Convert a color picture to grey-scale
  - Thresholding (terskling)

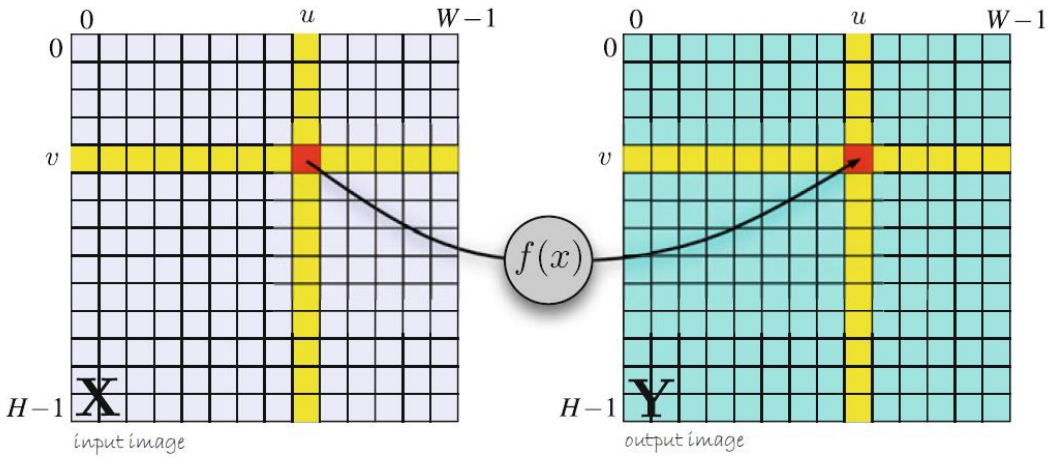
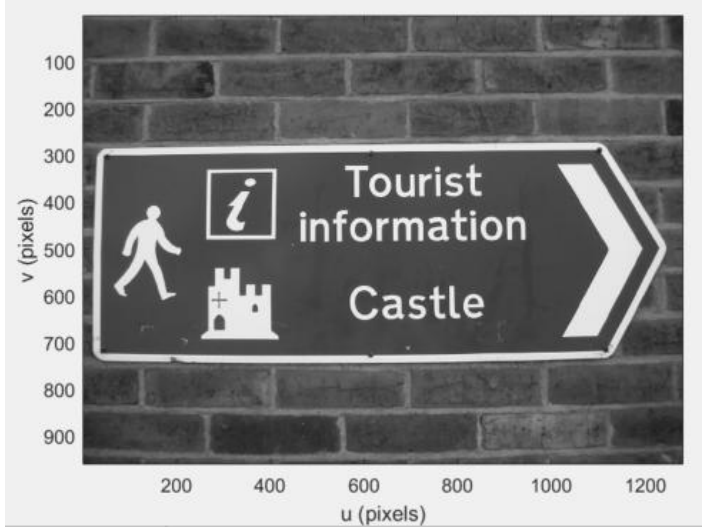
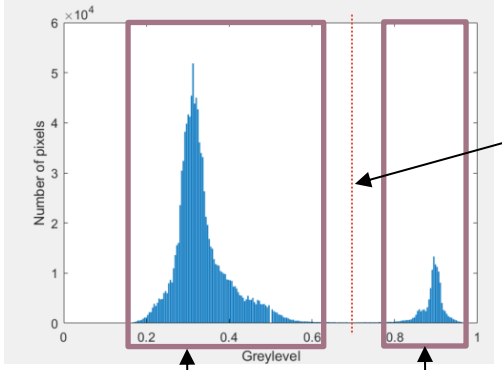


Fig. 11.8 Monadic image processing operations. Each output pixel is a function of the corresponding input pixel (shown in red)

Thresholding (terskling)

- Seperates pixels into two categories based on intensity

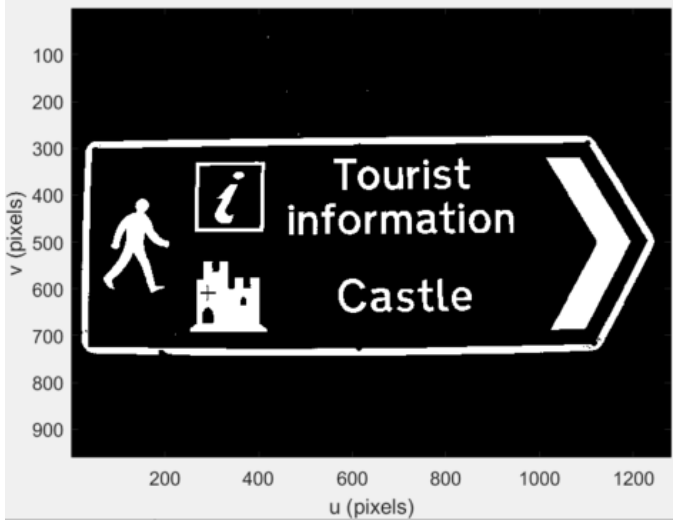




Threshold

Dark pixels

Light pixels

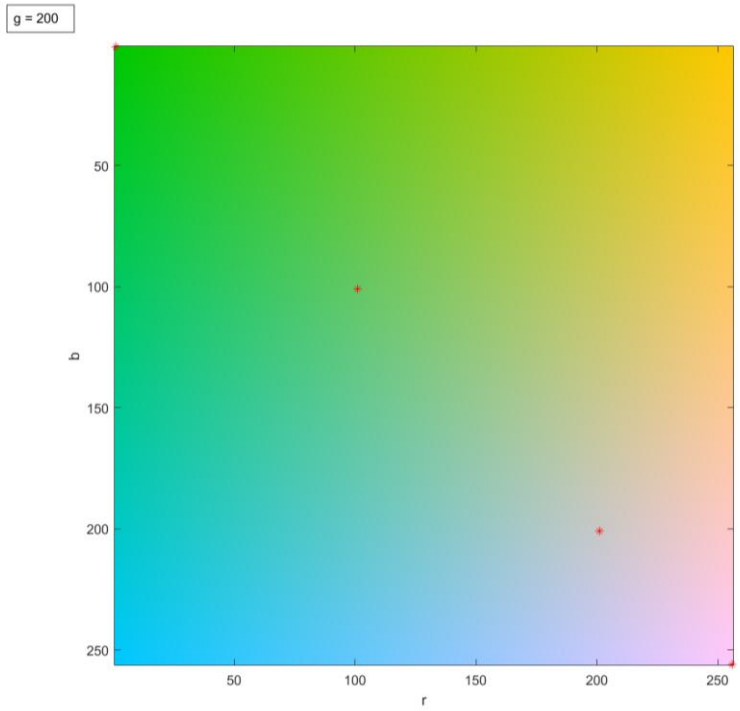


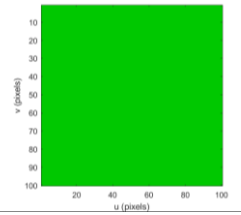
Segmentation and detection

- To divide the picture into meaningful regions
- For example, based on color

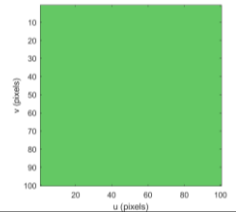
Color segmentation

- RGB-color space is typically not the most suitable one
- All pictures below have the same green value

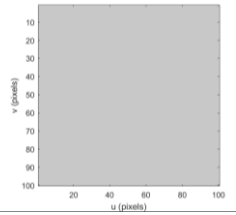




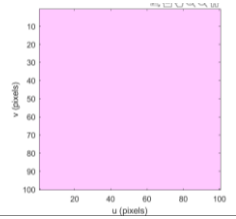
RGB(0, 200, 0)



RGB(100, 200, 100)



RGB(200, 200, 200)



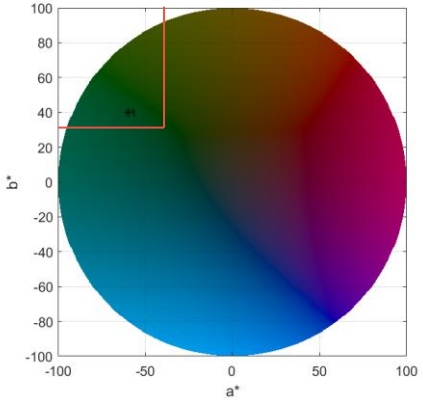
RGB(255, 200, 255)

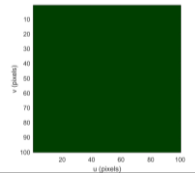
Color segmentation

L*a*b* is typically more suitable
Look at a* and b* for different L* values

```
>> showcolorspace('Lab', 'L', 20)
>> showcolorspace('Lab', 'L', 40)
>> showcolorspace('Lab', 'L', 60)
>> showcolorspace('Lab', 'L', 80)
```

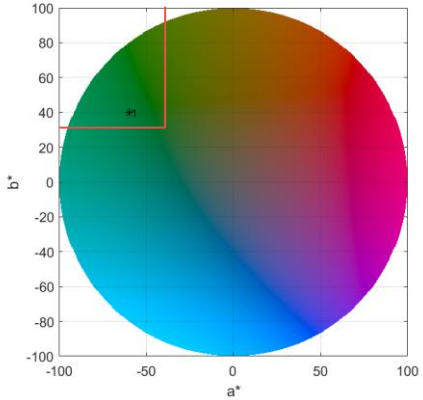
L=20

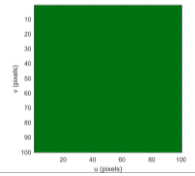




L*a*b*(20, -60, 40)
RGB(0, 63, 0)

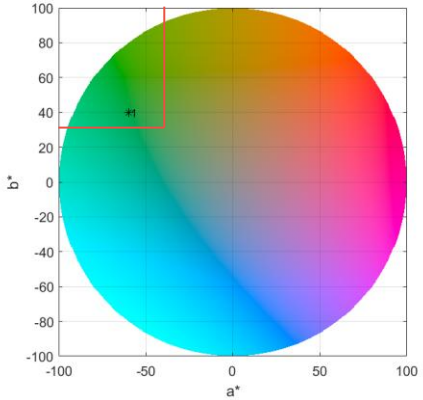
L=40

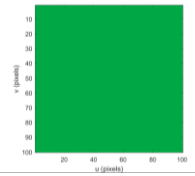




L*a*b*(40, -60, 40)
RGB(0, 114, 17)

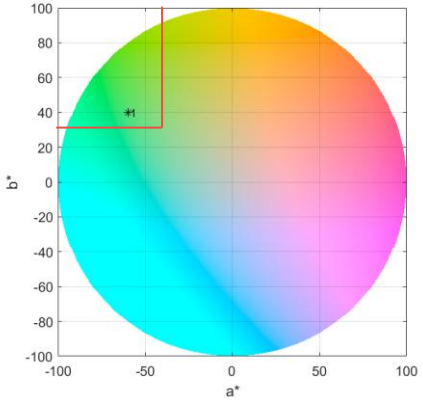
L=60

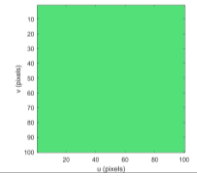




L*a*b*(60, -60, 40)
RGB(0, 168, 69)

L=80



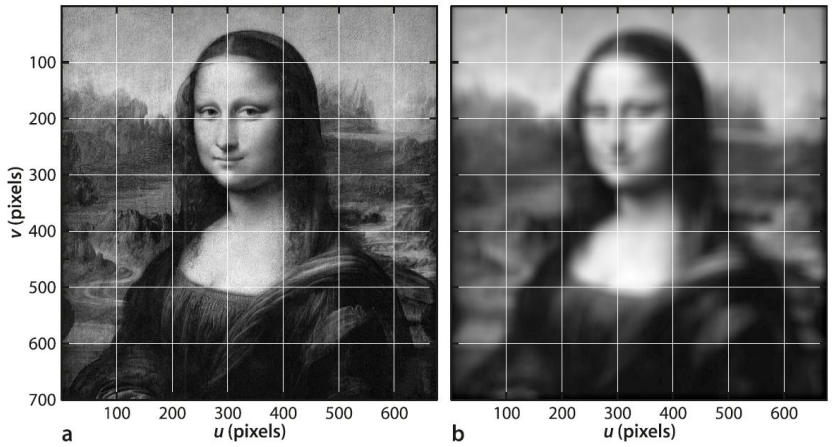


L*a*b*(80, -60, 40)
RGB(83, 224, 120)

11.5 Spatial operations (maskeoperasjoner)

- Each pixel in output image is a function of neighboring pixels to the pixel in original /input image
- Examples:
  - Edge detection (Kantdeteksjon)
  - Smoothing (Glatting)

Each pixel in output image is the average of pixel-values in a 21x21 rectanble around corresonding pixel in input image



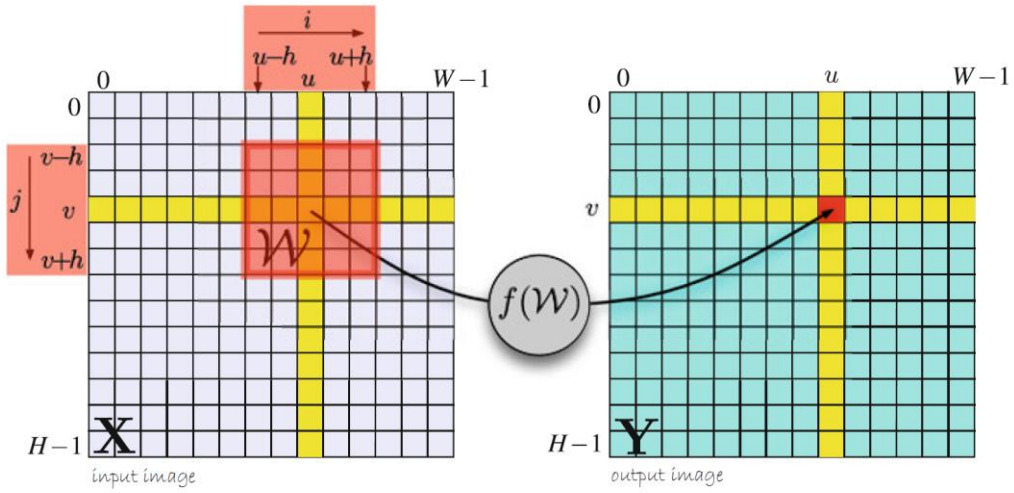


Fig. 11.13 Spatial image processing operations. The red shaded region shows the window $\mathcal{W}$ that is the set of pixels used to compute the output pixel (shown in red)

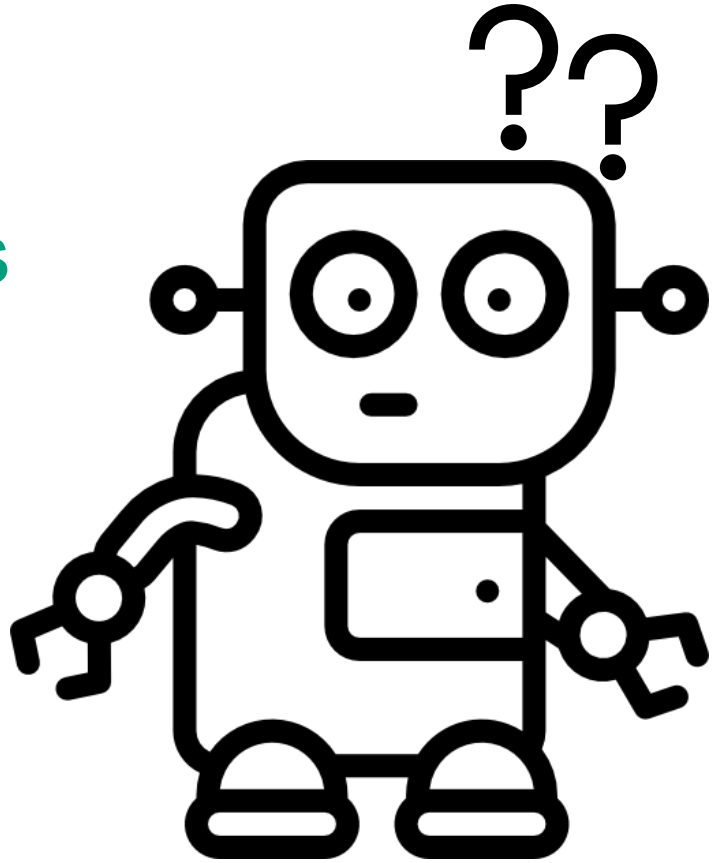
Related QUT Robot Academy video:

- Monadic image processing
  - https://robotacademy.net.au/masterclass/image-processing/?lesson=650
- Image thresholding
  - https://robotacademy.net.au/masterclass/image-processing/?lesson=655
- Separating intensity and color
  - https://robotacademy.net.au/masterclass/color/?lesson=726
- Finding color objects in a scene
  - https://robotacademy.net.au/masterclass/color/?lesson=727

Related QUT Robot Academy video:

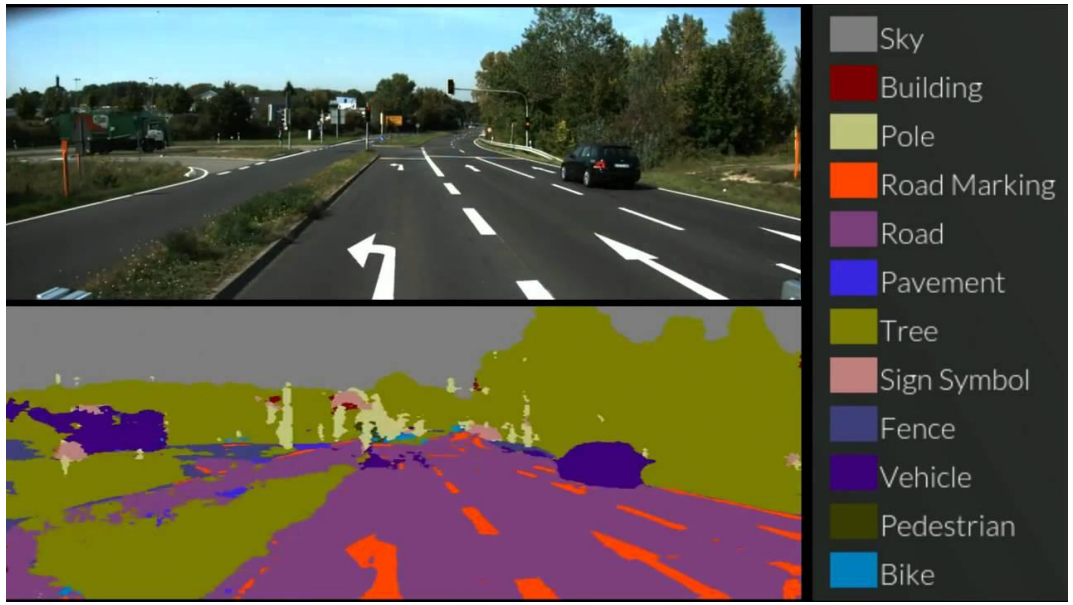
- Image region
  - https://robotacademy.net.au/lesson/image-region/
- Multiple image regions
  - https://robotacademy.net.au/lesson/multiple-image-regions/
- Spatial operators
  - https://robotacademy.net.au/masterclass/spatial-operators/?lesson=670
- Spatial operators - handling edge pixels
  - https://robotacademy.net.au/masterclass/spatial-operators/?lesson=671
- Finding edges
  - https://robotacademy.net.au/masterclass/spatial-operators/?lesson=674

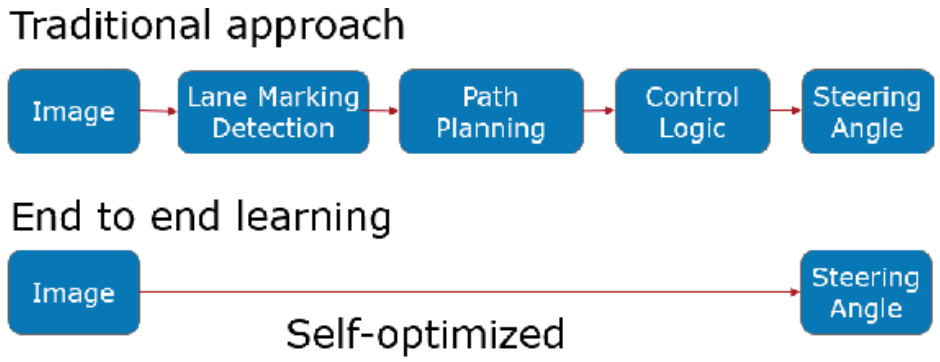
Machine learning in robotics



Machine Learning in Robotics

- Peter Corke's RVC provides a classical (bottom up) approach to robotics
- Last decade however, robotics has (also) undergone a paradigm shift; "AI/ML revolution"
- Used both for sensor/perception/vision as well as control purposes
- Question: Is control side more/less safety critical than perception side?





What is Machine Learning

"Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed"

— Arthur L. Samuel, AI pioneer, 1959

[probably the first, and undoubtedly the most popular definition]

(This is likely not an original quote but a paraphrased version of Samuel's sentence "Programming computers to learn from experience should eventually eliminate the need for much of this detailed programming effort.")

Arthur L Samuel. "Some studies in machine learning using the game of checkers". In: *IBM Journal of research and development* 3.3 (1959), pp. 210–229.

Venndiagram: AI, ML, DL, CNN, Generative AI, GPT, etc.

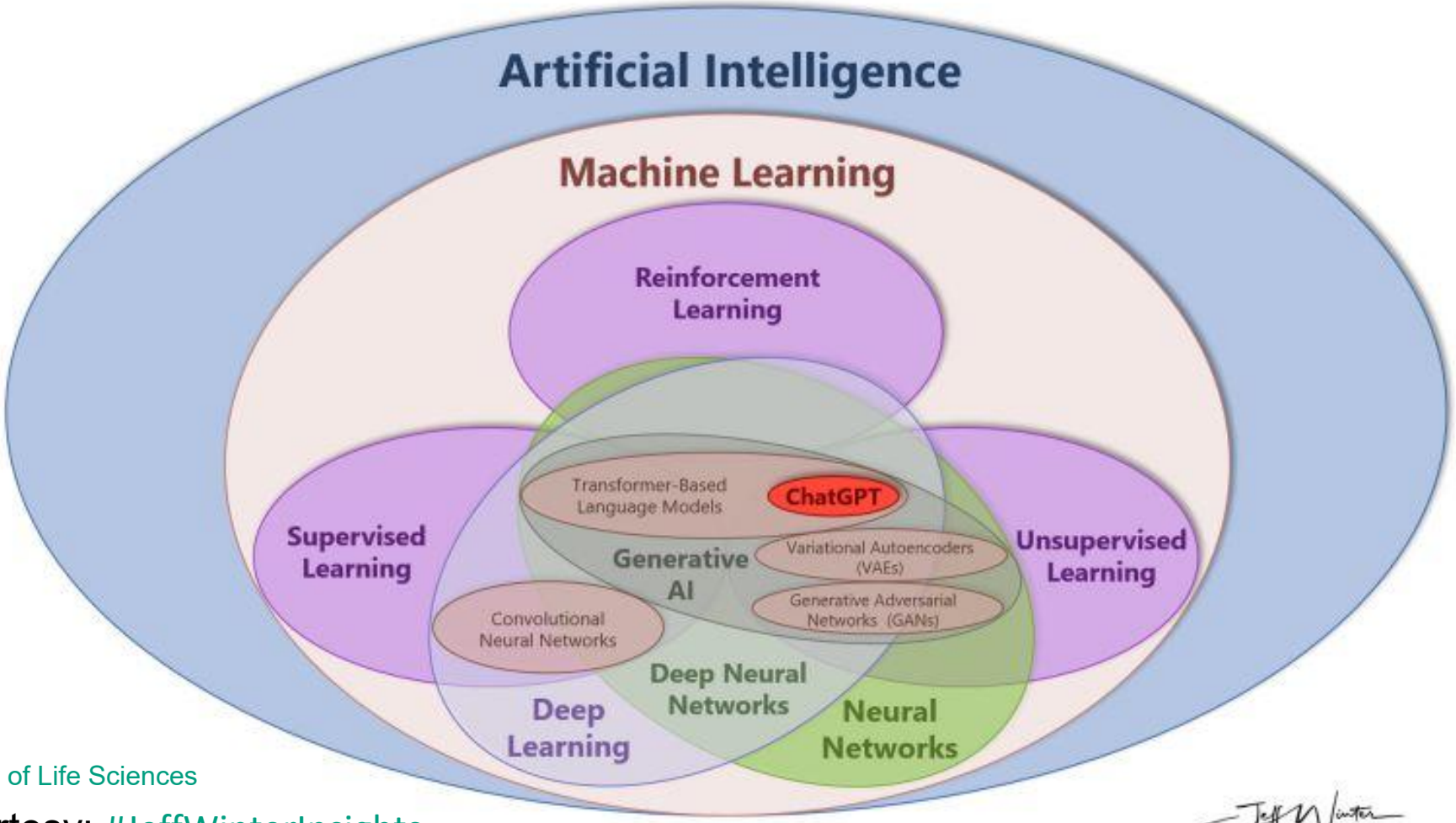


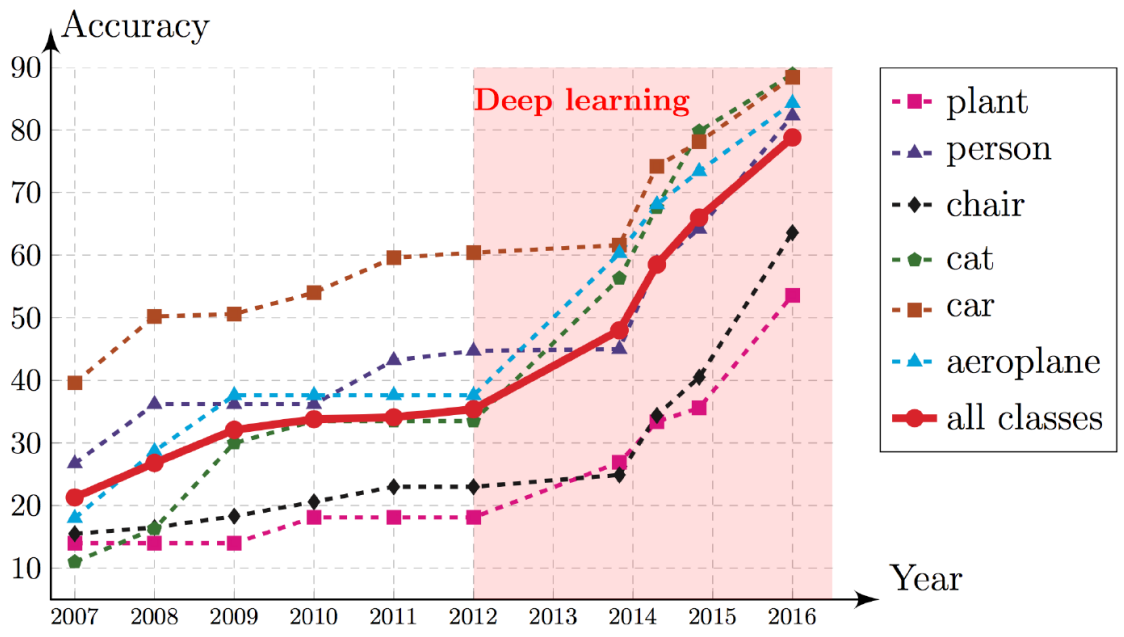
Photo courtesy: #JeffWinterInsights

Background – 3 broad categories of ML

Supervised Learning	Unsupervised Learning	Reinforcement Learning
• Labeled data • Direct feedback • Predict outcome/future	• No labels/targets • No feedback • Find hidden structure in data	• Decision process • Reward system • Learn series of actions

Convolutional Neural Network (CNN)

- 1950s-1970s: Perceptron – the first learning machine
- 1960s-1986: Backpropagation – Rumelhart et al.
- 1989: Convolutional network architecture – LeCun et al. → Hand written symbol recognition
- 1995: Check Reader – Bell Labs → Fielded in 1996, and used for 10-20% of all checks in the US in the early 2000s
- Mid-2000s – object classification getting decent results
- 2010-2012: Massive improvements in classification accuracy -> CNNs “ImageNet moment”!



Object detection in Pascal VOC 2007 Challenge

Large Language Model (LLM)

- -> 2010s Natural Language Processing (NLP) - supervised
- 2014 First capable generative models: variational auto-encoder & Generative Adverbial Network (GAN)
- 2017 Google Brain introduces Transformer-architecture
  - Attention is all you need! (NIPS2017)

Attention is **all you need**

A Vaswani, N Shazeer, N Parmar… - Advances in neural …, 2017 - proceedings.neurips.cc

… to attend to **all** positions in the decoder up to and including that position. **We need** to prevent

… **We** implement this inside of scaled dot-product **attention** by masking out (setting to −∞) …

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Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 13].

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

†Work performed while at Google Brain.

‡Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Large Language Model (LLM)

- -> 2010s Natural Language Processing (NLP) - supervised
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- 2017 Google Brain introduces Transformer-architecture
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- 2018 OpenAI introduce first semi-supervised Generative Pre-trained Transformer (GPT) network
  1. Unsupervised generative “pre-training”
  2. Supervised discriminative “fine-tuning”
- 2022- today: Generative AI, LLM “ChatGPT moment”!

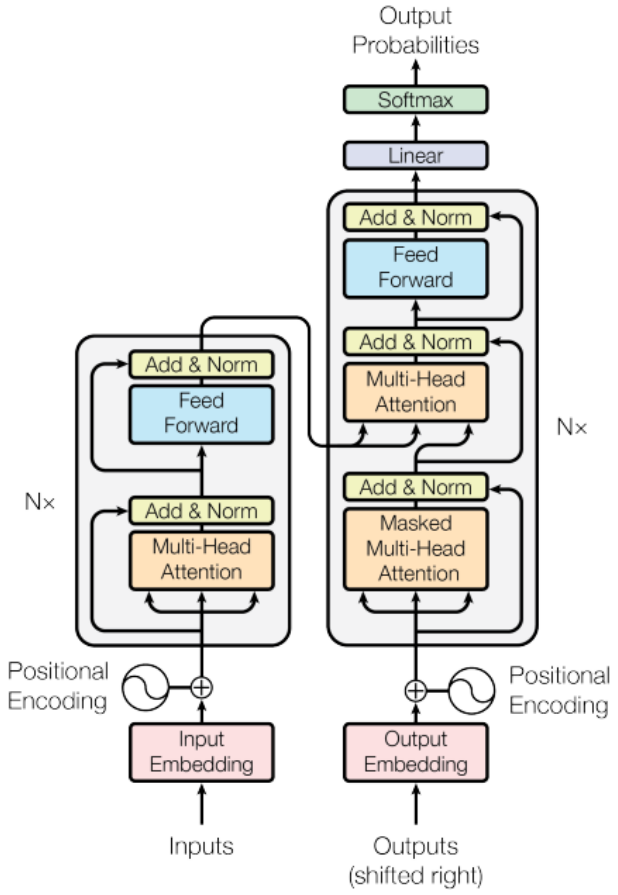


Figure 1: The Transformer - model architecture.

Home assignment: Lex Fridman podcast and questions

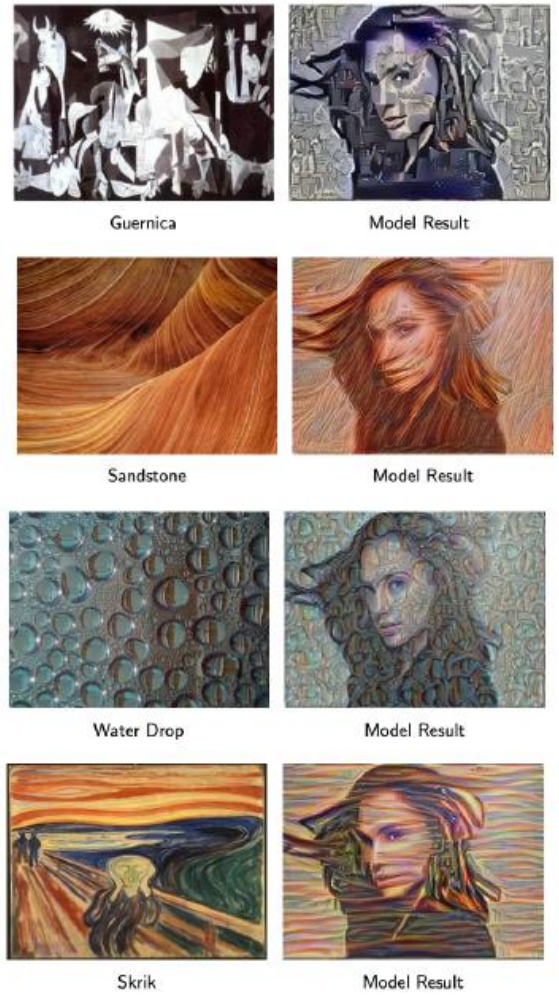
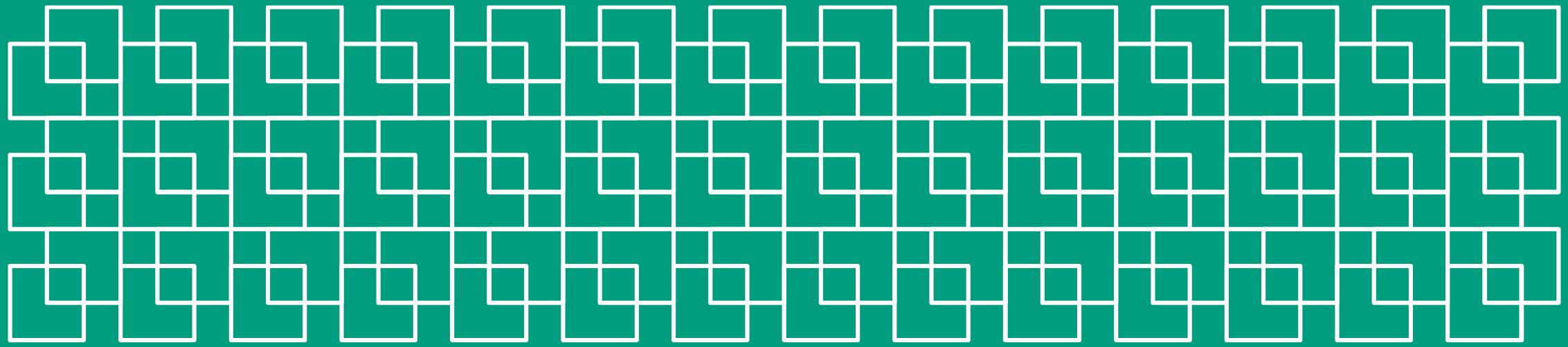
- Q1: What are the main differences and bottlenecks for ML within Robotics as compared to "regular" ML (that is not performed in the physical world, e.g., image classification).
- Q2: What does the "bitter lesson of machine learning" imply according to Rich Sutton?
- Q3: What counter-arguments can be used against Rich Sutton?

www.youtube.com › watch

Sergey Levine: Robotics and Machine Learning | Lex Fridman ...

Sergey Levine is a professor at Berkeley and a world-class researcher in **deep learning**, **reinforcement** ...

Jul 14, 2020 · Uploaded by Lex Fridman

Summary: ML in Robotics

- Last decade, robotics has (also) undergone a paradigm shift; “AI/ML revolution”
  - Feature/Classifier definitions by expert
  - End-to-end deep learning
- Robotics has some distinctive features which profound implications and limitations on ML:
  - Data collection (& labeling)
  - Reward/Cost function evaluation
  - Robustness, safety and reliability/assurance!